National 5 Engineering Science

Analogue Electronics

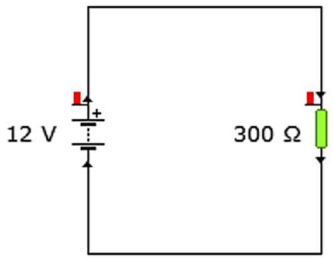
Section 1 — Attempt ALL questions

Total marks: 40

Name: _____ Class: _____ Date: _____

Question 1 (4 marks)

A resistor of 300 Ω is connected to a 12 V supply.



(a) Calculate the current flowing through the resistor. (2 marks)

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(b) State the formula used. (1 mark)

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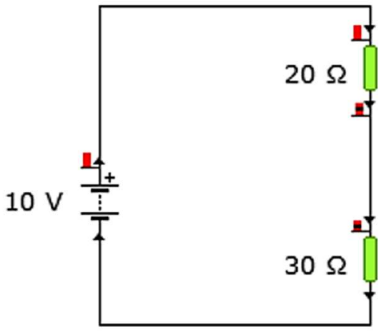
(c) State the unit of current. (1 mark)

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Question 2 (6 marks)

Two resistors of 20 Ω and 30 Ω are connected in series to a 10 V supply.



(a) Calculate the total resistance. (1 mark)

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(b) Calculate the circuit current. (2 marks)

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(c) Calculate the voltage across the 30 Ω resistor. (3 marks)

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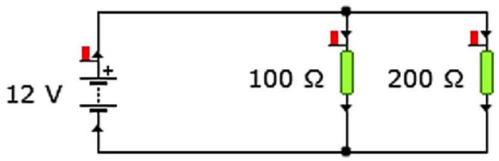
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Question 3 (6 marks)

Two resistors of 100 Ω and 200 Ω are connected in parallel to a 12 V supply.

(a) Calculate the total resistance. (3 marks)



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(b) Calculate the total circuit current. (2 marks)

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(c) State one characteristic of a parallel circuit. (1 mark)

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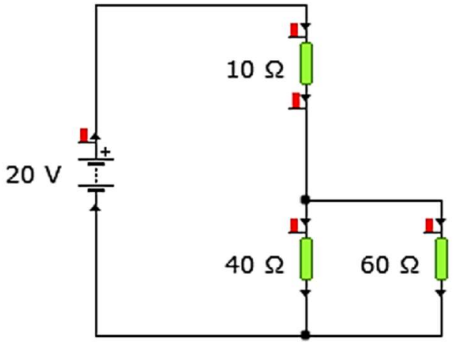
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Question 4 (5 marks)

A 10 Ω resistor is connected in series with two resistors (40 Ω and 60 Ω) in parallel.

The supply voltage is 20 V.

(a) Calculate the total resistance of the parallel section. (2 marks)



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(b) Calculate the total circuit resistance. (1 mark)

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(c) Calculate the circuit current. (2 marks)

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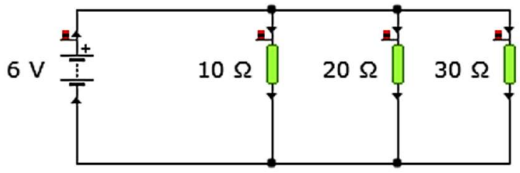
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Question 5 (5 marks)

Three resistors of 10 Ω, 20 Ω and 30 Ω are connected in parallel to a 6 V supply.

(a) Calculate the total resistance. (3 marks)



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(b) Calculate the current through the 10 Ω resistor. (2 marks)

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Question 6 (4 marks)

A student requires a resistance of 50 Ω but only has two 100 Ω resistors.

(a) Draw a circuit to achieve 50 Ω. (2 marks)

[Space for diagram]

(b) Prove by calculation that the total resistance is 50 Ω. (2 marks)

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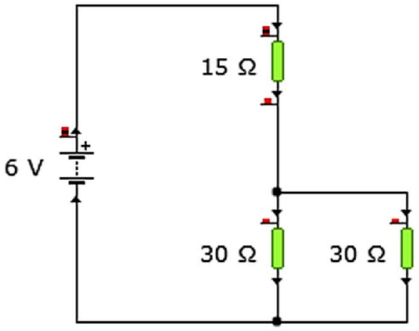
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Question 7 (5 marks)

A 6 V supply feeds a circuit where a 15 Ω resistor is connected in series with two 30 Ω resistors in parallel.

(a) Calculate the total resistance. (2 marks)



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(b) Calculate the circuit current. (1 mark)

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(c) Calculate the voltage across the parallel section. (2 marks)

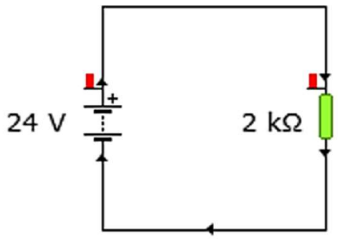
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Question 8 (5 marks)

A 24 V supply is connected to a 2 kΩ resistor.



(a) Calculate the current in milliamps (mA). (3 marks)

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(b) The resistor is replaced with a 1 kΩ resistor.

State what happens to the current and explain why. (2 marks)

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